SOVEREIGN DEBT

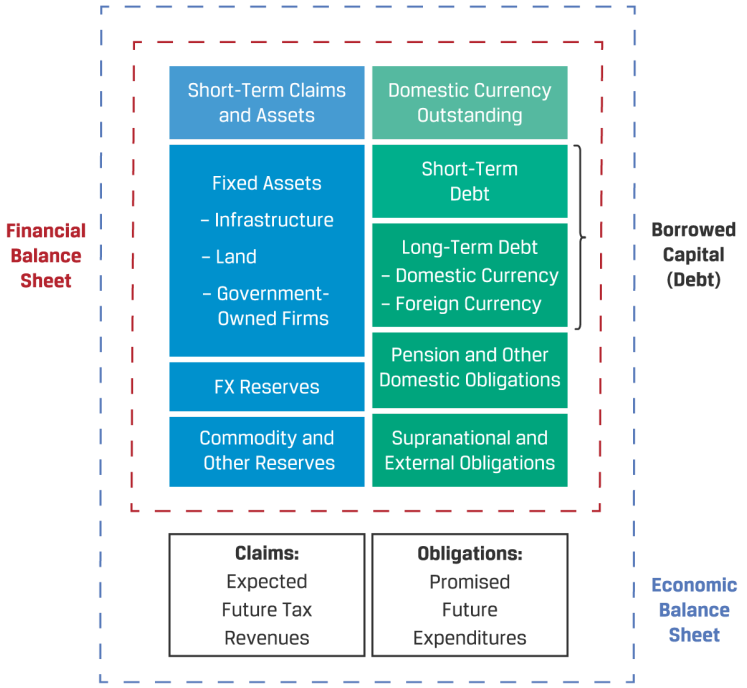
Learning Outcome

- describe funding choices by sovereign and non-sovereign governments, quasi-government entities, and supranational agencies

National or sovereign government issuers are distinguished by their legal authority to establish and maintain a country's public goods and services and their ability to tax economic activity in their jurisdiction. Additional sources of repayment for their debt obligations include tariffs, usage fees, and cash flows from government-owned enterprises. The size and scope of public goods and services provided by national, versus regional or local, governments vary widely among markets, as does the degree of government involvement in the economy. Similar to private sector issuers, a government's "economic balance sheet" may be used to illustrate the sources and uses of funding, as shown in Exhibit 1, which also includes activities of the central bank.

Exhibit 1:

Government "Balance Sheet"



▶ Expand / collapse Extended Description

While private issuers prepare and file periodic financial statements in accordance with generally accepted accounting principles (GAAP), public sector financial accounting standards vary widely and are often prepared using cash, rather than accrual-based, principles, typically excluding such items as the depreciation of fixed public goods, such as federal highways, or the accrual of unfunded liabilities, such as government pension obligations. In contrast, the economic balance sheet in Exhibit 1 includes expected future claims and obligations and is therefore of greater relevance for public versus private sector issuers.

government itself, quasi-government agencies, and local governments also differs among countries. As a result, this gives rise to non-sovereign issuers based in the same jurisdiction as the sovereign issuer, a topic discussed in Lesson 3.

A key distinction among national government issuers is the difference between developed market and emerging market sovereign issuers.

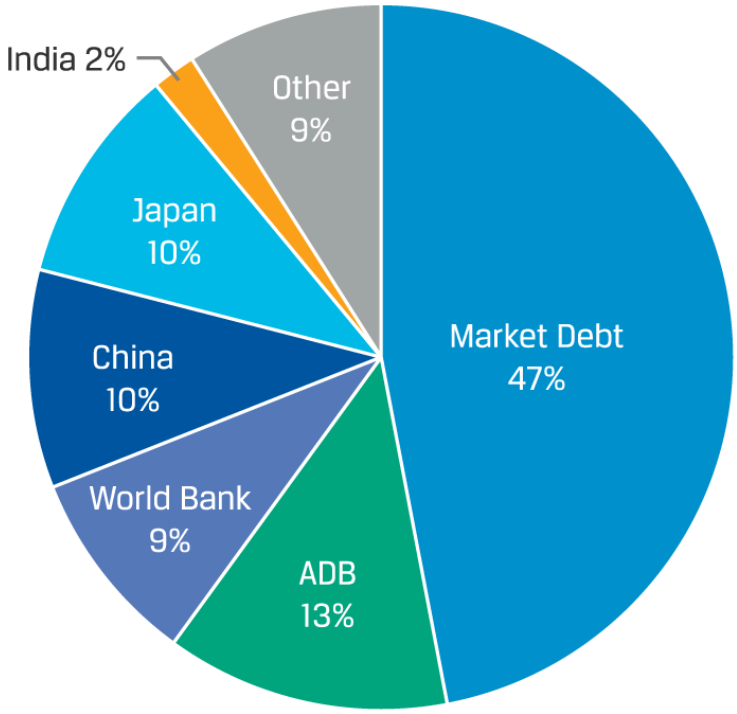
- *Developed market (DM) sovereign issuers:* DMs are characterized by a strong, stable, well-diversified domestic economy. DM national government budgets primarily comprise consistent, recurrent outlays financed with broad-based individual and business tax cash flows, resulting in stable and transparent fiscal policy. DM fixed-income securities are denominated in a major currency commonly held in reserve by foreign governments. These features allow DM sovereign governments to issue what is often referred to as default-risk-free debt with unconstrained market access across the maturity spectrum.
- *Emerging market (EM) sovereign issuers:* EMs are usually characterized by higher growth but less stable and less well-diversified economies subject to greater fluctuations over the economic cycle. EMs often depend on a dominant domestic industry or industries, such as commodities, and may involve more state-owned or state-controlled enterprises. Central government budget priorities may involve investments to expand economic and social infrastructure that exceed current domestic tax cash flows, giving rise to external or supranational funding. EM sovereign debt securities are often denominated in a **restricted domestic currency**, or one with limited convertibility into other currencies due to illiquidity. Currency restrictions may limit foreign investment and constrain access to longer-term maturities in domestic currency.

In the case of emerging market sovereign issuers, it is important to further distinguish between *domestic* debt issued in the domestic currency and **external debt**, or debt owed to foreign creditors. Sovereign bonds issued in domestic currency are often held mostly by domestic financial institutions and other domestic investors.

External debt of emerging and frontier market sovereign issuers comprises debt from supranational financial organizations and debt issuance denominated in a foreign currency held by foreign private investors. Earlier modules cited examples of emerging market sovereign Eurobond issuance, such as Romania's euro-denominated bond and the quasi-government PT Indonesia Infrastructure Finance US dollar–denominated bond. Investors based in developed markets who purchase these bonds do not face the same *direct* currency risk as for domestic currency emerging market sovereign debt. However, they face *indirect* exposure to currency fluctuations, because their returns depend on an issuer's ability to generate foreign currency revenue through international capital, goods, and services flows sufficient to meet foreign currency interest and principal payments. The following example highlights the case of the government of Sri Lanka's external debt.

Government of Sri Lanka External Debt

In May 2022, Sri Lanka became the first Asia-Pacific country in many years to default on its external debt when it failed to make USD78 million in foreign currency interest payments following a 30-day grace period. Sri Lanka's government debt rose sharply in the years preceding the default due in large part to infrastructure projects financed by global debt markets, supranational agencies, and major trading partners. Total government debt outstanding rose from 87% of Sri Lanka's GDP to over 100% in 2020, while both its economy and foreign currency reserves shrank during that year due to a collapse in tourism and declining remittances from citizens working abroad. Higher inflation and commodity prices in the wake of the 2022 war in Ukraine compounded the country's external debt problems and ultimately led to default.



Note: "Market Debt" is primarily international sovereign bonds denominated in US dollars and owned by foreign private investors, including mutual funds.
Sources: Government of Sri Lanka, Department of External Resources (www.erd.gov.lk). Lilian Karunungan and Amelia Pollard, "Sri Lanka Falls into Default for the First Time Ever," Bloomberg (19 May 2022). Bank for International Settlements, "Debt Securities Statistics." "Infrastructure Financing in Sri Lanka: Lessons Learnt and Future Collaborations," European Institute for Asian Studies Policy Brief 16/2021.

A nation's fiscal policy determines the *level* of sovereign debt through central government spending, including both budgetary requirements and debt service costs, against tax receipts, fees, and revenue from government-owned enterprises or other sources. Government budget deficits increase the need for borrowing while surpluses reduce borrowing needs. Forecasts of sovereign debt levels must consider not only changes in fiscal policy but also the sensitivity of expenditures and revenues to changes in economic growth and inflation.

Government debt management policies address the *composition* of sovereign debt—that is, short term versus long term, as well as other features. Sovereign debt issues include the following:

- Short-term securities (with maturities ranging from 1 to 12 months), often known as Treasury bills, which are usually zero-coupon instruments sold at a discount to par.
- Medium- and long-term securities often known as Treasury notes and bonds. Fixed-rate coupon instruments in the domestic currency are most common, but sovereigns also issue floating-rate, inflation-linked, and foreign currency instruments.
- While not acting as the issuer, some sovereign governments guarantee certain other instruments that, in effect, make them a type of sovereign debt. The most prominent example is mortgage-backed securities, especially in the United States, that conform to certain criteria. Such instruments will be discussed in detail later.

Given the sovereign right to levy taxes on economic activity, national governments represent the lowest default risk among domestic currency issuers and are usually the largest bond issuer in a domestic market.

Recall from earlier in the corporate issuers reading that the Modigliani–Miller (MM) theorem stated that under certain strict conditions, a company's choice of capital structure is irrelevant in determining the present value of the firm's expected future cash flows. In the case of sovereign government debt, the Ricardian equivalence theorem introduced in an earlier economics lesson, which was based on similar simplified assumptions, gives an

Under Ricardian equivalence, taxpayers expect government debt to be offset by higher future taxes, suggesting that a sovereign government should be indifferent between collecting taxes today or raising debt of any maturity based on the following assumptions:

- Taxpayers smooth consumption over time, saving expected future taxes today for future payment.
- Taxpayers form rational expectations that today's tax cuts will result in future tax increases.
- Capital markets are perfect with no transaction costs, and taxpayers can freely borrow and lend.
- Taxpayers are altruistic on an intergenerational basis; that is, they pass on tax savings to descendants.

The relaxation of these Ricardian equivalence assumptions to allow imperfect capital markets and friction costs between bond investors and taxpayers gives rise to debt management policies that offer liquidity benefits to investors and other issuers across the maturity spectrum.

For example, short-term government issuance provides investors with a high degree of liquidity and safety and often substitutes for bank deposits. Under normal market conditions, the prices of these very short-term government securities involve a liquidity premium; that is, their yields-to-maturity will tend to be *lower* than they otherwise would be in the absence of this liquidity benefit. While short-term sovereign debt is typically issued at regular intervals (discussed later) and standard maturities, government issuers have a high degree of flexibility to alter the size of these periodic auctions to manage uncertain cash flows, such as near-term tax receipts or unforeseen funding needs.

Strict adherence to Ricardian equivalence assumptions suggests that governments should fund themselves with the shortest possible maturity to minimize borrowing costs and avoid paying any term premium associated with longer-term interest rates. However, similar to the example of corporate bonds in the prior module, excessive government reliance on short-term funding introduces rollover risk, or uncertain refinancing costs, increasing the variability of budget costs and therefore tax rates. Since taxpayers are unable to perfectly smooth consumption over time nor do they form rational expectations about current versus future taxes across generations, fiscal instability creates uncertainty in future tax rates and threatens the stability of economic growth over time. Therefore, in practice, governments seek to minimize interest rate and rollover risks by distributing debt across maturities, while issuing debt in regular, predictable intervals.

Fiscal policies differ significantly across the 38 OECD member countries, but their sovereign debt maturity structures have shown far less variance. At the end of 2021, the weighted average maturity across the OECD was just over one year higher than the average at the end of 2007. Long-term debt represented 85% of total sovereign debt outstanding, largely unchanged since 2007, while annual issuance was evenly split between short and long term (short-term debt must be issued more frequently because it matures sooner). This time frame includes two major recessions and significant sovereign debt issuance.

Figure 2:

Average Time to Maturity of OECD Nations' Sovereign Debt Outstanding (Years)

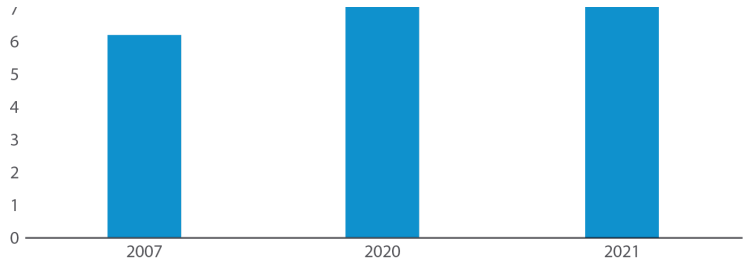
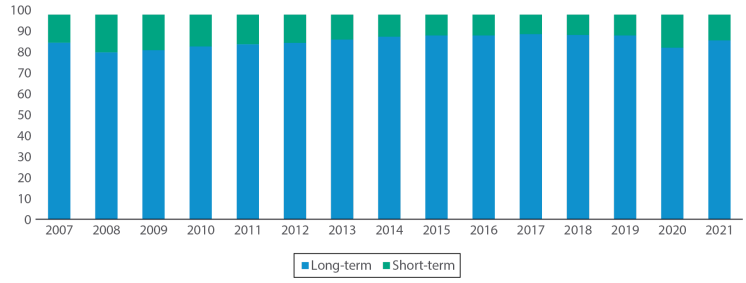


Figure 3:

Maturity Structure of OECD Nations' Sovereign Government Debt Outstanding (Percent)



Source: OECD, "OECD Sovereign Borrowing Outlook 2022" (2022). https://doi.org/10.1787/b2d85ea7-en.

The use of medium- and long-term sovereign funding issuance must balance the trade-off of higher borrowing costs with greater fiscal stability, while considering liquidity or other benefits of issuing bonds with the lowest default risk across the maturity spectrum.

The benefits of issuing sufficient longer-term sovereign government securities to maintain liquidity across maturities include the following:

- *Establishment of a risk-free benchmark for all debt of specific maturities:* The first fixed-income module demonstrated the use of sovereign bond yields to calculate an issuer's credit risk premium. Government debt policies often mandate the regular issuance of benchmark securities across maturities to increase capital market efficiency and transparency for private issuers.
- *Use in managing and hedging market interest rate risk:* Market participants such as financial intermediaries and asset managers use medium- to long-term government securities and related derivatives to manage interest rate risk separately from credit risk.
- *Preferred use as collateral in repo and derivative transactions:* Longer-term government securities are the most common form of collateral for both repo and derivative transactions due to their high degree of liquidity and safety. Recall from the prior learning module that repurchase agreements offer security sellers a means of financing and offer security buyers a short-term collateralized investment with relatively little liquidity or default risk.
- *Government bond use in monetary policy and foreign exchange reserves:* Recall from an earlier lesson that central banks use government securities in fulfilling their monetary policy mandate, which includes the outright purchase and sale of securities and repo contracts with government bonds as collateral. Given their high degree of liquidity and safety, foreign market participants often hold their foreign currency reserves in the form of liquid foreign government bonds.

While sovereign governments seek to maintain regular and predictable bond market access across maturities, changes in the amount of debt, the level of deficits, and the difference between short- and long-term interest rates lead to changes in debt management policy, as shown in the following example.

In October 2001, the United States stopped issuing 30-year Treasury bonds due to cost concerns related to paying interest on the notes. At the time, Treasury officials saw a reduced need for future borrowing because the U government was posting a budget surplus. The move stunned the markets and resulted in the largest one-day price move in years due to supply concerns. In prior years, the Treasury Department suspended the three- and seven-year issues. The 10-year Treasury became the de facto benchmark for measuring the performance of long-term bonds during the suspension of the 30-year bond. Issuance of the 30-year US Treasury bond resumed in February 2006 to cover deficits created by tax cuts and war spending.

Sources: Wall Street Journal, "Yields Plunge after Treasury Halts Issue of 30-Year Bonds" (31 October 2001). www.wsj.com/articles/SB1004542963185681400
TSC Staff, "Treasury to Resume 30-Year Bond on Feb. 9," TheStreet (2 November 2005). www.thestreet.com/markets/rates-and-bonds/treasury-to-resume-30-year-bond-on-feb-9-10250971

Question Set

1. Identify the following statement as true or false. Justify your answer.

Emerging market external sovereign debt is denominated in a foreign currency.

Hide Solution

Solution:

False. Although EM sovereigns do issue debt in foreign currencies to increase demand by foreign investors who desire less currency risk, many EM sovereigns have external debt denominated in their domestic currency as well, especially those with currencies that have exhibited exchange rate stability.

2. US-based investors of US dollar–denominated external sovereign debt can still face exposure to currency risk because:

A. sovereign fixed-income securities are denominated in freely floating domestic currencies.
B. the issuer's credit rating may be downgraded.
C. the issuer must be able to generate foreign currency revenue.

Hide Solution

Solution:

C is correct. Investors based in developed markets who purchase external debt of emerging and frontier market sovereign issuers face *indirect* exposure to currency fluctuations, because their returns depend on an issuer's ability to generate foreign currency revenue to meet foreign currency interest and principal payments through international capital, goods, and services flows.

3. Explain why national governments represent the lowest default risk among domestic currency issuers.

Hide Solution

Solution:

National governments are distinguished by their sovereign right to generate tax cash flows from economic activity within a jurisdiction, unlike private sector issuers that rely on operating cash flows and secondary sources of repayment, such as asset sales, to service debt obligations.

4. Due to their high degree of liquidity and safety, the most common form of collateral for both repo and derivative transactions is:

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

C. longer-term government securities.

Hide Solution

Solution:

C is correct. Longer-term government securities are the most common form of collateral for both repo and derivative transactions due to their high degree of liquidity and safety.

5. A sovereign government faced with growing deficits and concerns about rollover risk would *most likely*:

A. lower tax rates.
B. stop servicing the debt.
C. issue medium- to long-term securities.

Hide Solution

Solution:

C is correct. Government budget deficits increase the need for borrowing at a higher level while surpluses reduce borrowing needs. Excessive government reliance on short-term funding introduces rollover risk, or uncertain refinancing costs, increasing the variability of budget costs and therefore tax rates. Therefore, in practice, governments seek to minimize interest rate and rollover risks by distributing debt across maturities while issuing debt in regular, predictable intervals. A is incorrect because taxes should be raised, not lowered, in such a scenario. B is incorrect because such an action would put the government in default, with severe consequences for its credit standing and its ability to meet fiscal policy commitments.

Discuss Filter

Discussion

What do you want to discuss... (required)

LW For Q4 in Question Set, why option A is incorrect?

Created 5 months ago by Ling Wang 2 replies | Last Activity: 3 months ago

Show All replies Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search